Uber NYC

Pickup Data Analysis

January – June 2015

Exploratory Data Analysis Report

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Tools: Python · Pandas · Plotly · Folium

Dataset: 100,000-record NYC Uber sample (H1 2015)

1. Executive Summary

This report presents an exploratory data analysis of Uber pickup activity in New York City during the first half of 2015. Using a 100,000-record sample from the uber-raw-data-janjune-15.csv dataset, the analysis examines temporal demand patterns, dispatching base performance, airport-specific demand, and geospatial pickup distribution across the city.

Key Findings

- **Growing demand:** Pickups grow steadily from January through June — May and June are the busiest months.
- **Weekend dominance:** Saturday and Friday consistently record the highest pickup volumes across all months.
- **Evening peak:** Peak hourly demand occurs between 6 PM and midnight; the lowest activity is 3–5 AM.
- **Concentrated supply:** Dispatching base B02764 dominates, accounting for approximately 40% of all trips.
- **Airport split:** LGA (LaGuardia) sees higher and more volatile demand than JFK throughout the day.

2. Dataset Overview

Attribute	Detail
Source File	uber-raw-data-janjune-15_sample.csv
Records	100,000 rows
Columns	Dispatching_base_num, Pickup_date, Affiliated_base_num, locationID
Period Covered	January 2015 – June 2015
Derived Features	month, day (day of week), hour

Feature Engineering

The Pickup_date datetime column was parsed to extract three new features:

- **month** — Calendar month name (January–June)
- **day** — Day of the week (Monday–Sunday)
- **hour** — Hour of the day (0–23)

No null values were found in the core columns. The dataset was clean and required minimal preprocessing before analysis.

3. Monthly & Day-of-Week Pickup Analysis

A grouped bar chart was constructed using a cross-tabulation of month vs. day of week, giving the total number of pickups for each month–day combination.

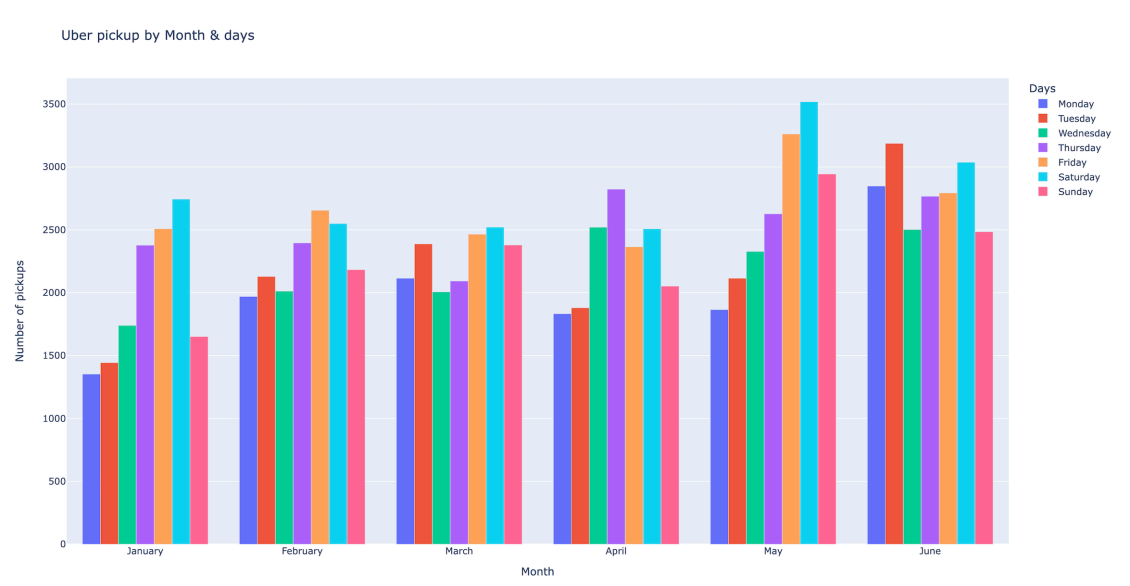

Chart 1: Uber Pickup by Month & Days (Grouped Bar Chart)

Month	Mon	Tue	Wed	Thu	Fri	Sat	Sun
January	1,353	1,444	1,740	2,378	2,508	2,745	1,651
February	1,970	2,129	2,013	2,396	2,655	2,550	2,183
March	2,115	2,388	2,007	2,093	2,465	2,522	2,379
April	1,833	1,880	2,521	2,823	2,365	2,508	2,052
May	1,865	2,115	2,328	2,627	3,262	3,519	2,944
June	2,848	3,187	2,503	2,767	2,793	3,037	2,485

Observations

- **January** has the lowest counts across all days — post-holiday slowdown and winter weather.
- Pickup volume grows month-over-month, peaking in May and June.
- In **May**, Saturday recorded the highest single-day figure of approximately **3,519 pickups**, followed by Friday (~3,262).

- **Monday** consistently records the lowest pickup count, reflecting lighter activity at the start of the week.
- Overall growth from January to June represents roughly a 60–70% increase across most days.

Key Insight: Weekends (Saturday & Sunday) drive the most demand in Spring months. Seasonal growth is significant — demand in June is nearly double that of January on Mondays.

4. Hourly Rush Analysis by Day of Week

Trip counts were grouped by day of week and hour of day to reveal how ride-hailing demand changes over a 24-hour cycle. The resulting multi-line chart plots each day as a separate series.

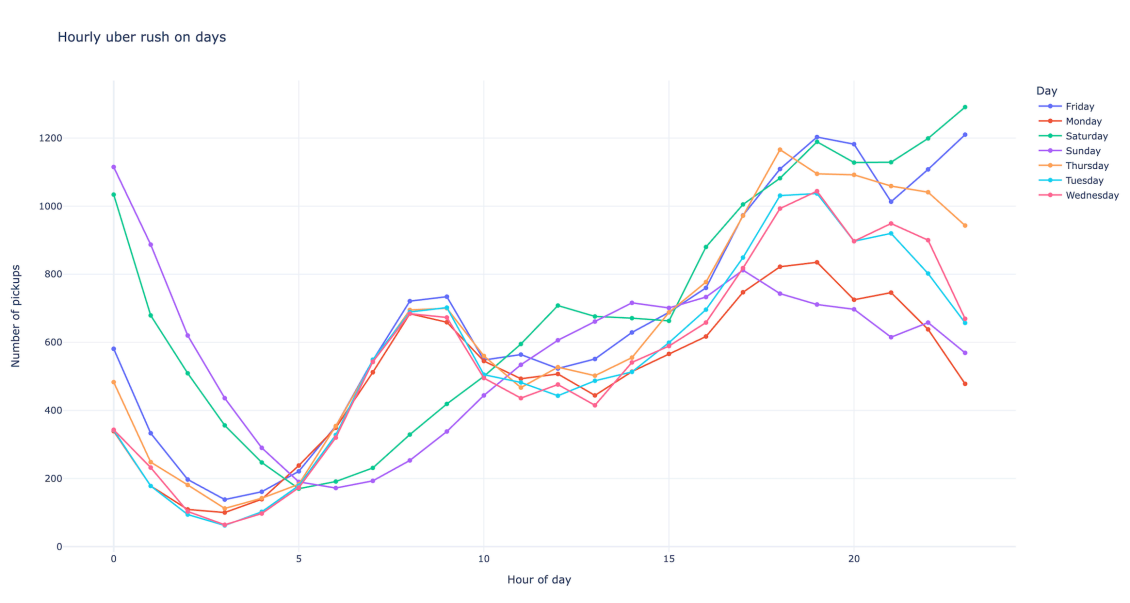

Chart 2: Hourly Uber Rush on Days (Multi-Line Chart)

Observations

- All days follow a consistent **bimodal pattern**: a morning shoulder peak (~8–9 AM) and a dominant evening peak starting around 5–6 PM continuing until midnight.
- Activity hits its lowest point between **3 AM and 5 AM** on all days.
- **Saturday and Sunday** show elevated demand from midnight onward, reflecting late-night entertainment trips.
- **Saturday at hour 23** records the single highest pickup count — over 1,300 pickups.
- **Monday** has the flattest evening peak (~800–850 pickups), consistent with it being a lighter travel day.

- A mid-morning dip (~10–11 AM) is visible across all days, followed by an afternoon recovery.

Key Insight: Driver allocation should be maximised between 6 PM and 1 AM, especially on Friday and Saturday. Early morning hours (3–5 AM) represent ideal driver rest windows.

5. Dispatching Base Performance (Pareto)

A Pareto chart was constructed with trip counts per dispatching base shown as bars (left axis) and cumulative percentage as a line overlay (right axis). The 80% threshold is marked with a dashed reference line.

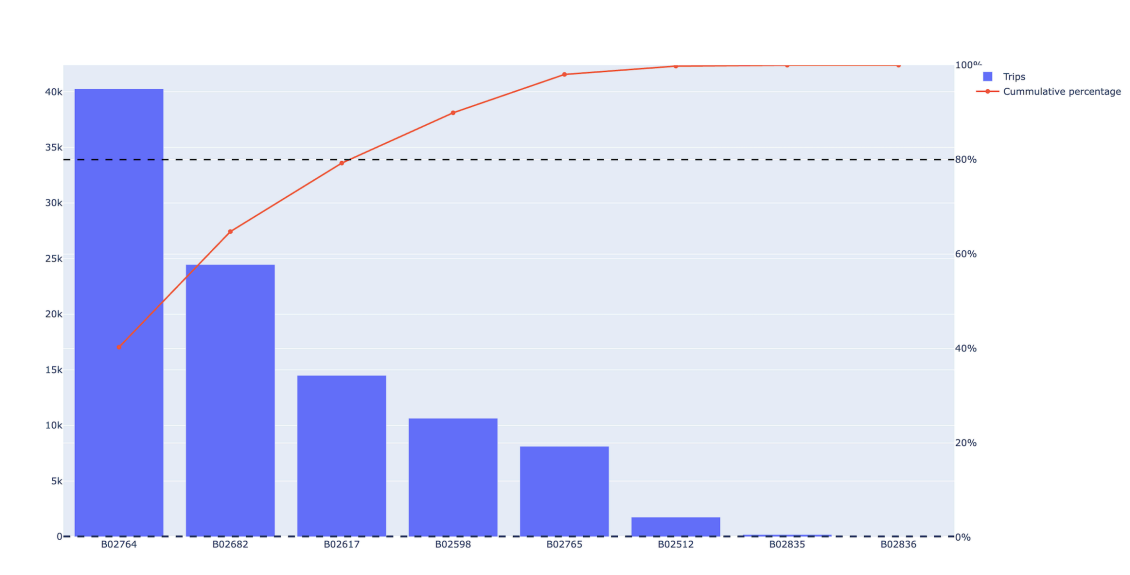

Chart 3: Pareto — Trips by Dispatching Base

Dispatching Base	Trips (approx.)	Cumulative %
B02764	~40,000	~40%
B02682	~24,000	~64%
B02617	~14,500	~79%
B02598	~10,500	~90%
B02765	~8,200	~98%
B02512	~1,800	~99.8%

Dispatching Base	Trips (approx.)	Cumulative %
B02835	~100	~99.9%
B02836	~50	~100%

Observations

- **B02764** is the dominant dispatching base with ~40,000 trips, comprising roughly 40% of total volume.
- The top 2 bases (B02764 + B02682) account for approximately 65% of all trips.
- Adding B02617 brings the cumulative share to ~80%, confirming the classic **Pareto Principle (80/20 rule)**.
- **B02512, B02835, and B02836** contribute negligibly — under 2% collectively.

Key Insight: Uber's NYC operations in H1 2015 were highly concentrated. Optimisation efforts targeting the top 3 bases would cover 80% of all rides.

6. Airport Demand Analysis: JFK vs LGA

An area chart compares the hourly trip volume at JFK (John F. Kennedy International) and LGA (LaGuardia) airports throughout the day. Airport trips were isolated using the relevant TLC location IDs.

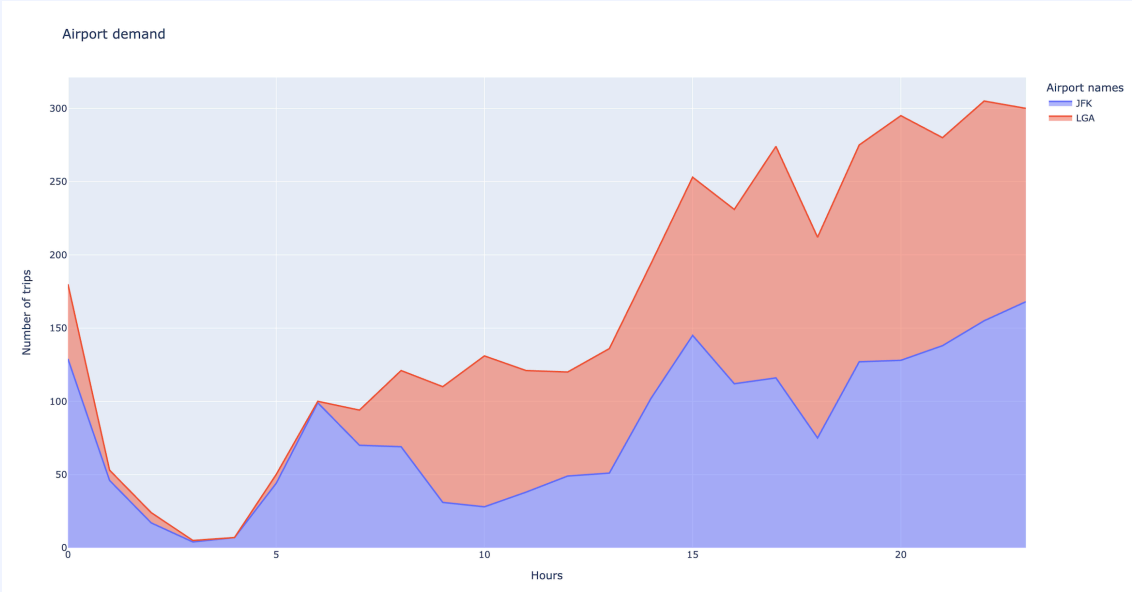

Chart 4: Airport Demand — JFK vs LGA (Area Chart by Hour)

Observations

- Both airports show a clear early-morning demand spike (midnight–1 AM), dropping to near-zero around 3–4 AM.
- **LGA consistently outperforms JFK** throughout most of the day: ~180 trips at midnight vs JFK's ~100.
- The gap widens significantly in the afternoon; by hour 22, LGA reaches **300+ trips** vs JFK's ~165.
- **LGA demand** is more volatile (more peaks and troughs); JFK shows a smoother, steadier climb.
- The two airports converge during the early-morning trough (hours 3–4).

> **Key Insight:** LGA is the busier Uber airport hub, likely due to its domestic-heavy traffic and proximity to Manhattan. JFK may represent fewer but longer, higher-value trips — warranting different surge pricing strategies.

7. Geospatial Analysis (Folium Heatmap)

A time-lapse heatmap was generated using Folium's HeatMapWithTime plugin, centred on New York City (40.7128N, 74.0060W) at zoom level 11. Pickup coordinates were grouped by hour, producing 24 hourly frames.

- High-density zones are concentrated in Midtown Manhattan, the Financial District, and around major transit hubs.
- The animation illustrates how pickup hotspots shift geographically throughout the day — from residential zones in the morning to nightlife corridors at night.
- The heatmap is rendered as an interactive HTML file, best viewed in a browser by running the notebook.

8. Summary of Key Findings

Analysis Area	Key Finding
Busiest Month	May (highest Saturday count: ~3,519 pickups)
Busiest Day	Saturday, followed by Friday
Peak Hours	6 PM – 1 AM
Slowest Hours	3 AM – 5 AM
Top Dispatching Base	B02764 (~40% of all trips)
80% Trip Concentration	Top 3 bases: B02764, B02682, B02617

Analysis Area	Key Finding
Busier Airport	LGA outpaces JFK for most hours of the day
Seasonal Growth	~60–70% increase in pickups from January to June

9. Tools & Libraries

Library	Purpose
pandas	Data loading, cleaning, feature engineering, aggregation
numpy	Numerical operations
plotly.express	Interactive bar, line, and area charts
folium	Geospatial time-lapse heatmap (HeatMapWithTime)
matplotlib	Supplementary static bar charts
seaborn	Styling and palette utilities

10. Conclusion

This analysis reveals clear and actionable patterns in Uber NYC demand during H1 2015. Temporal patterns strongly suggest demand is driven by commuting rhythms on weekdays and entertainment or leisure activity on weekends. The concentration of trips in a small number of dispatching bases and the dominance of evening hours offer concrete implications for fleet management and dynamic pricing strategies.

The airport demand analysis highlights LaGuardia as the primary airport hub, with distinct hourly patterns compared to JFK. The geospatial heatmap further confirms that demand hotspots shift meaningfully across the city throughout the day.

Potential Next Steps

- Incorporate full geo-coordinates for spatial clustering (K-means / DBSCAN).
- Merge with the complete dataset for statistical significance testing.
- Extend analysis to include surge pricing data and trip distance/duration metrics.
- Build a predictive demand model (time-series forecasting) by zone and hour.
- Compare H1 2015 patterns against 2014 data to quantify year-over-year growth.

Report generated from Uber_Data_Analysis.ipynb | Author: Alekhya Ayinam | May 2026